



中国科学院大学
University of Chinese Academy of Sciences

Design and Development of a 3D Printed Self-Solving Rubik's Cube Robot

Intelligence Software Engineer Course Project

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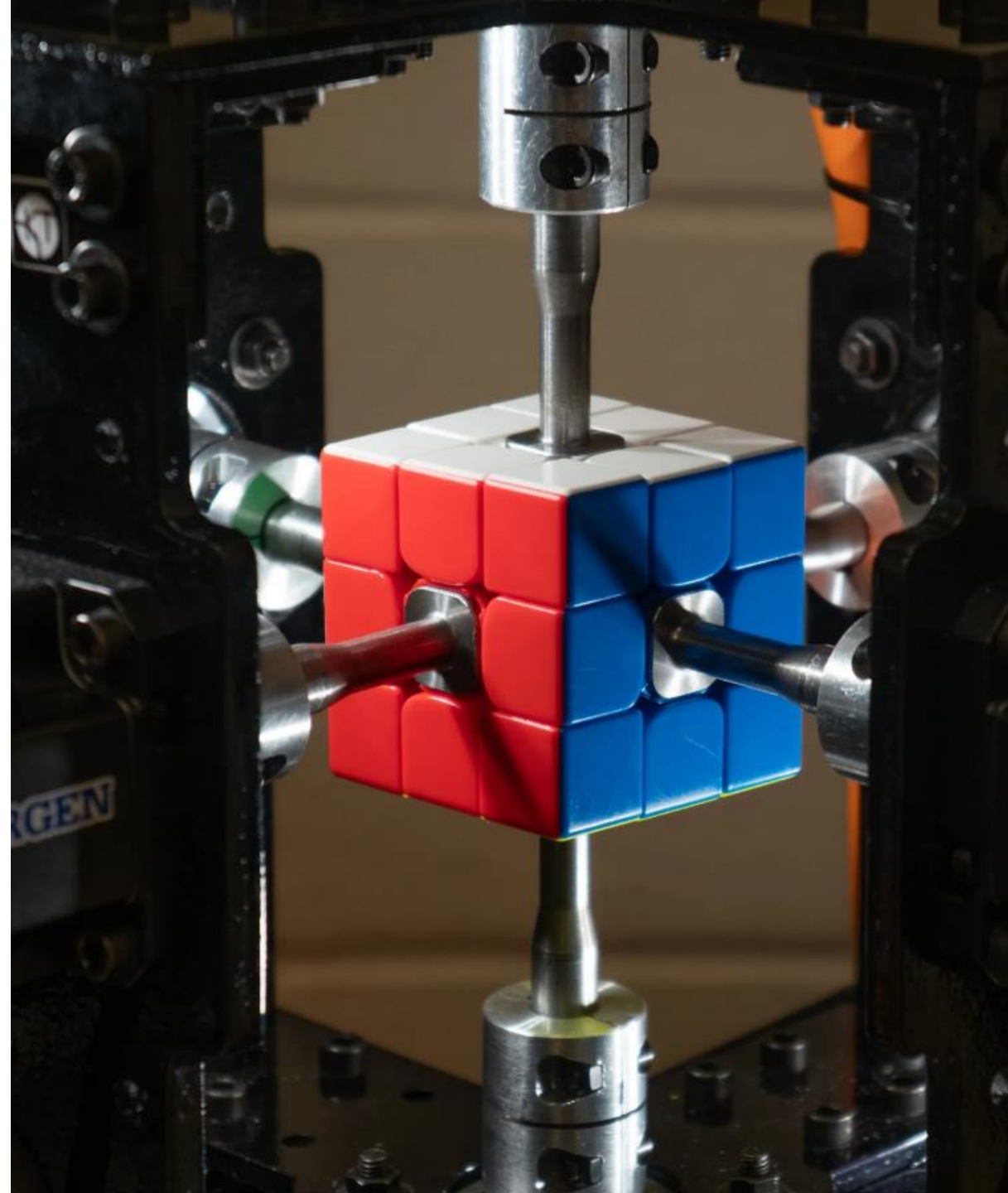
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1. Introduction

1. This project transcends the traditional task of puzzle-solving by framing the Rubik's Cube as a complex mechanical environment for Autonomous Sensorimotor Discovery. Rather than a mere toy, the cube is defined as a high-dimensional state-space system where a robotic agent must learn to navigate through physical interactions.
2. Our framework utilizes a custom-engineered 3D-printed apparatus equipped with four stepper motors, serving as the Experimental Apparatus. Controlled by an Arduino Mega 2560, the system facilitates a self-supervised learning loop where the robot executes low-level action primitives to observe and model state transitions.
3. This approach demonstrates the integration of affordable additive manufacturing with advanced robotics, shifting the focus from "hard-coded solving" to "grounded physical modeling." The ultimate goal is to validate a low-cost prototype capable of mapping unknown transition functions in a physical environment.



2. Formal Problem Definition

- The Rubik's Cube is not just a puzzle, but a physical system with formal mathematical structure:

The Rubik's Cube as a Formal State-Space System

- Mathematical Framework: State Space: The set of all possible physical configurations of the cube (approx. 4.3×10^{19} states).

Symbol	Meaning In Our System	Symbol	Meaning In Our System	Symbol	Meaning In Our System
S	State Space	All possible color configurations ($\approx 4.3 \times 10^{19}$ states)	S	State Space	All possible color configurations ($\approx 4.3 \times 10^{19}$ states)
A	Action Space		A	{U, U', U ² , R, R', R ² , F, F', F ² , L, L', L ² , D, D', D ² , B, B', B ² }	
T	Transition Function	Physical result of applying action \$a\$ in state (s)	T	Transition Function	Physical result of applying action \$a\$ in state \$s\$
s-0	Initial State		s-0	Scrambled cube configuration	
s-g	Goal State		s-g	Solved cube (all faces uniform)	

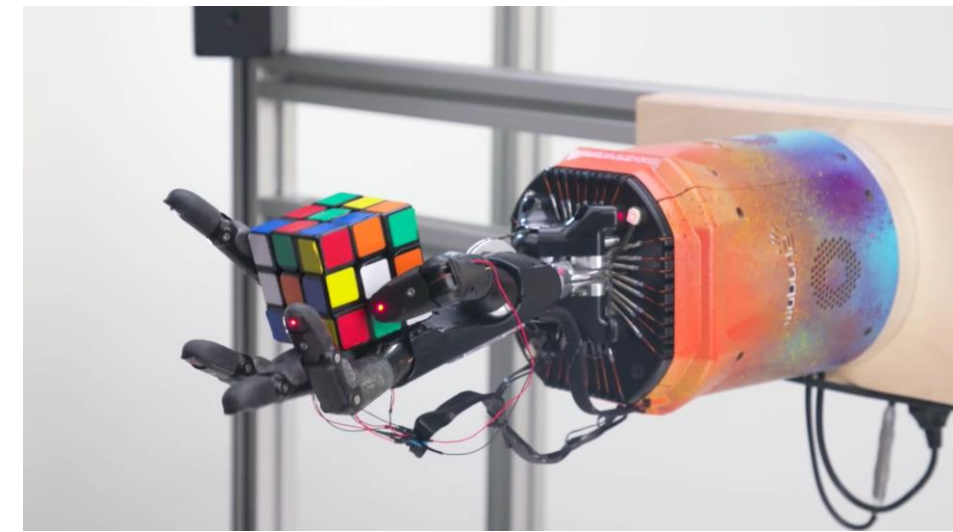


Fig.1. Different Self solving robot

Equation 1: System Dynamics $s_{t+1} = T(s_t, a_t)$

Where (T) is the physical transition function learned through experimentation.

$$\hat{T}_{\theta}(s, a) \approx T(s, a)$$

Equation 2: Learning Objective

Find parameters (theta) that approximate the true physics through robotic interaction

Transition Function (T): The unknown physical mapping where $T: S \times A \rightarrow S$.

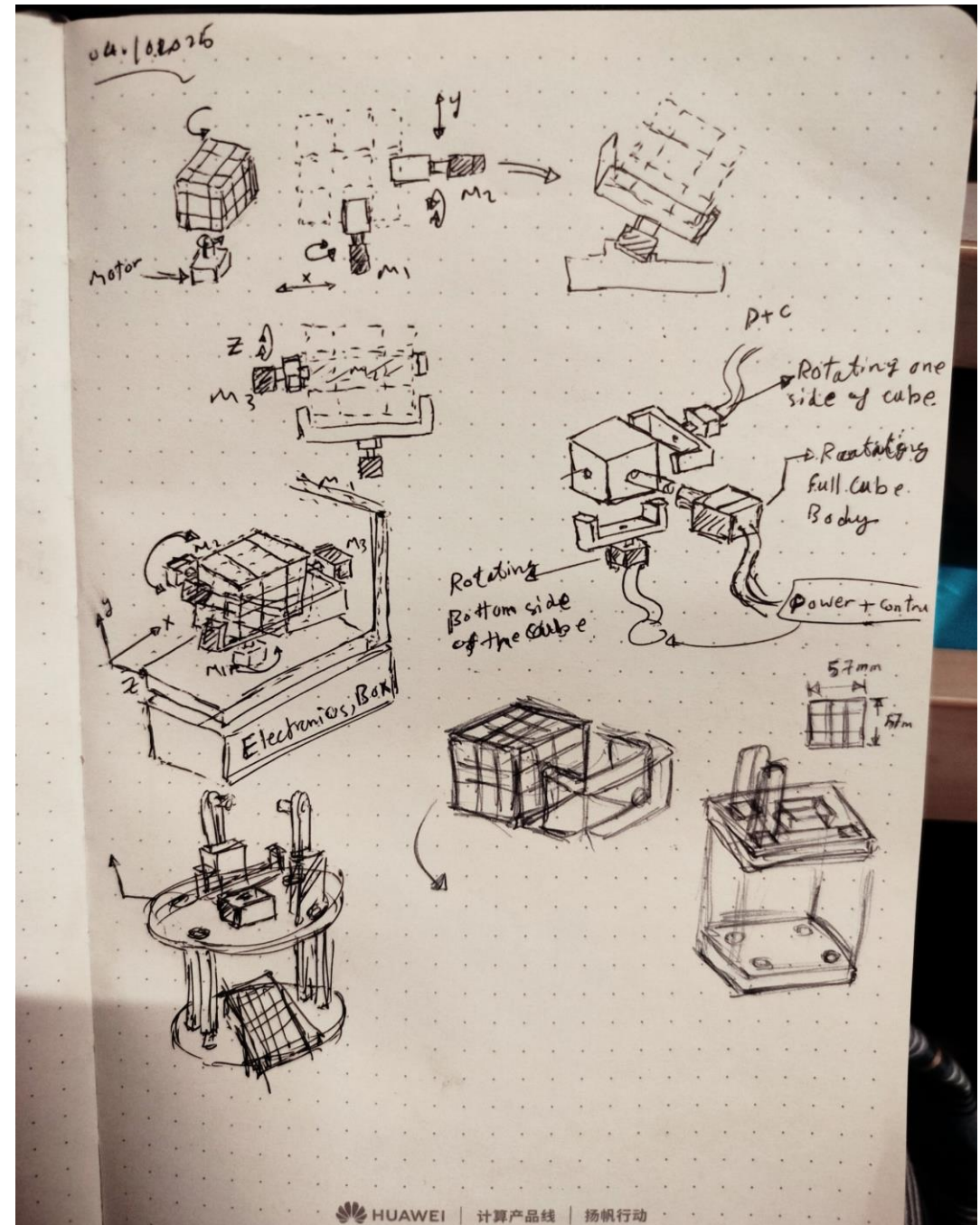
The Learning Objective: To train a neural network model M that approximates the transition function T :

$$M(s_t, a_t) \approx s_{t+1}.$$



Fig.2. Mathematics Rubik's Cube

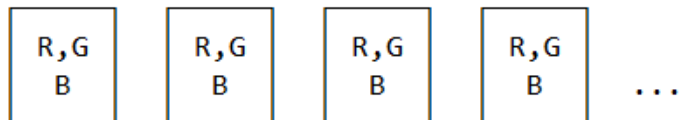
3. Solution





ROBOTIC DISCOVERY PLATFORM

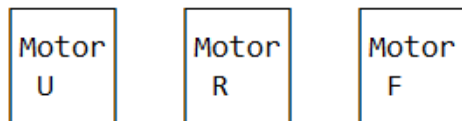
STATE OBSERVATION MODULE



54-dimensional observation vector

↑
| (serial data)

ACTION EXECUTION MODULE



3 degrees of freedom control

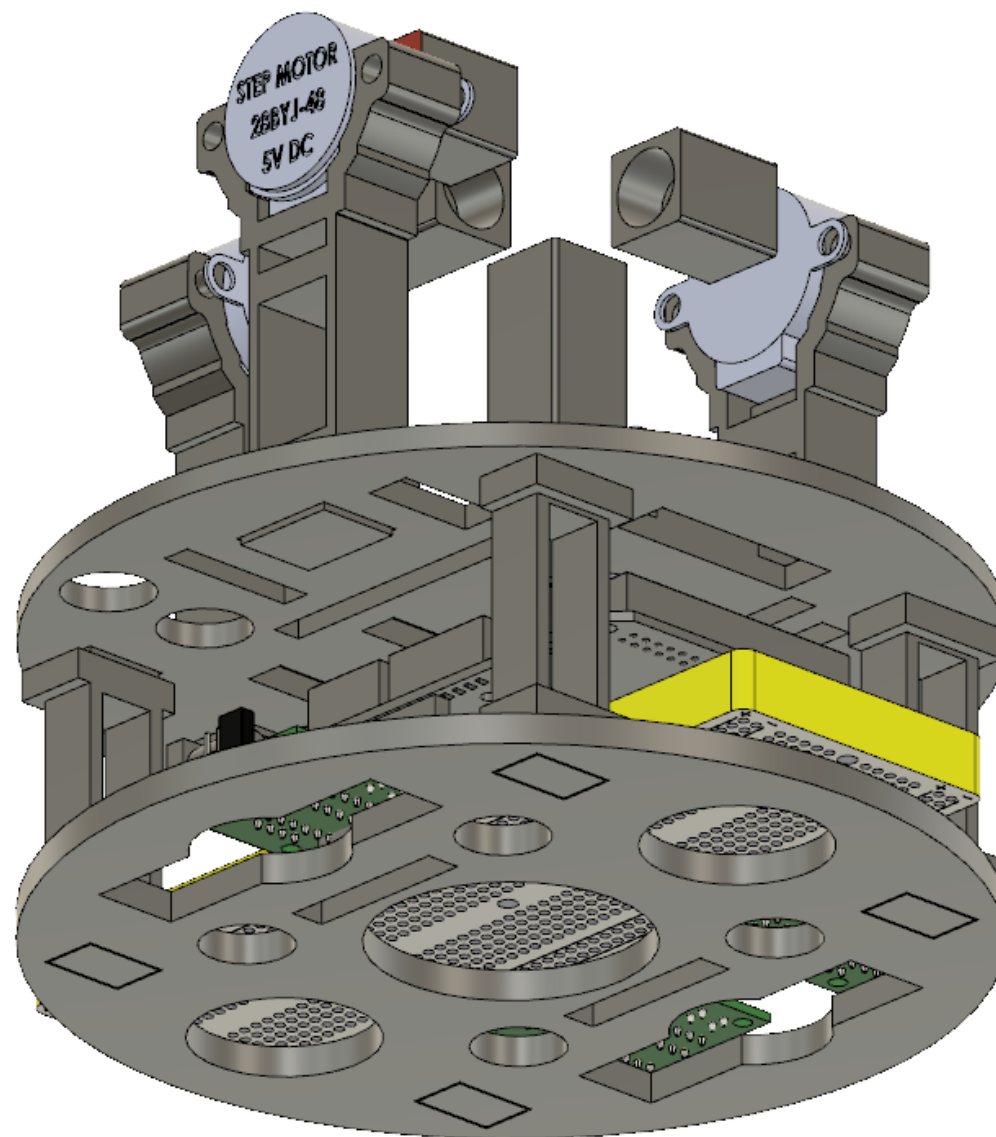
↑
| (pulse signals)

CONTROL & LOGGING SYSTEM

Arduino Mega 2560

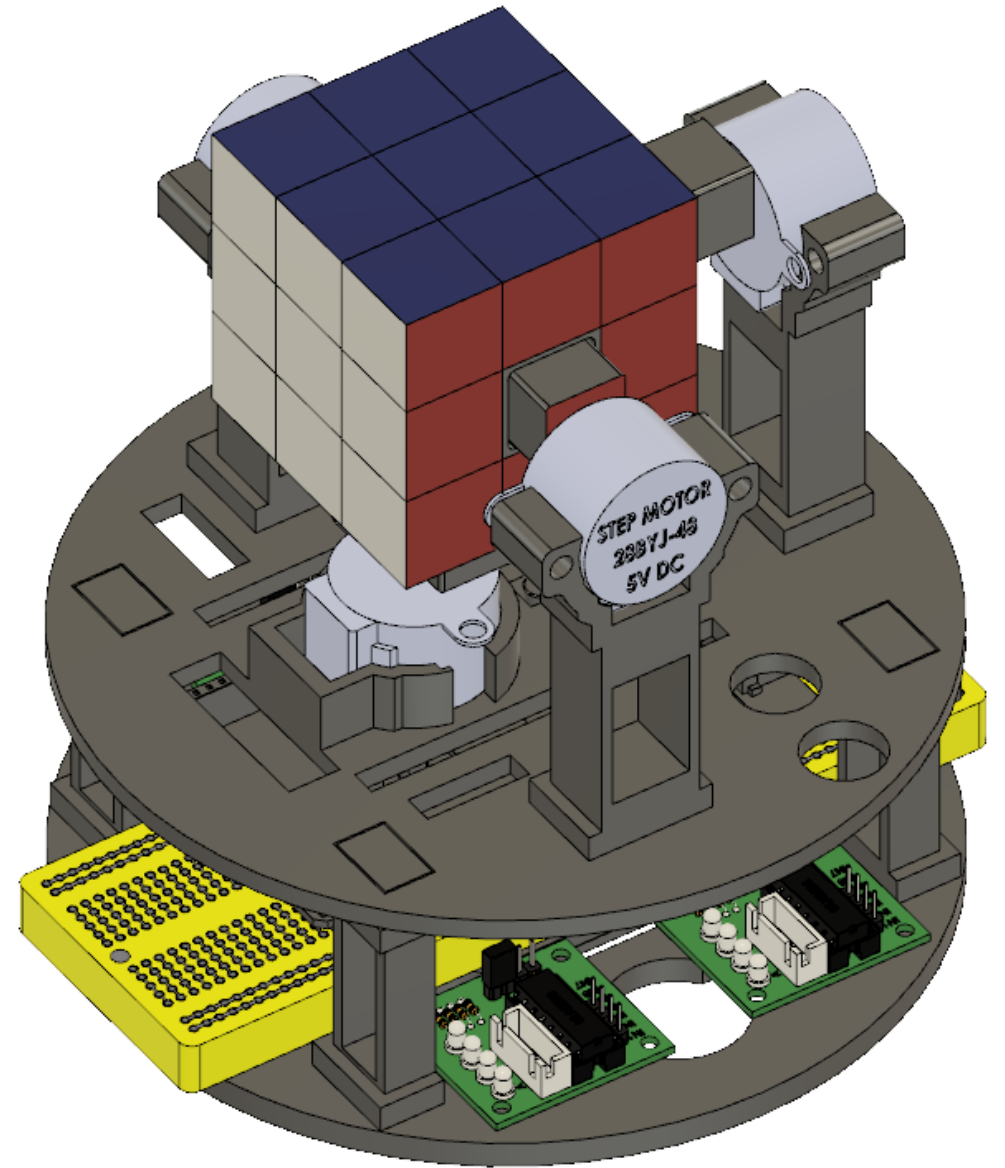
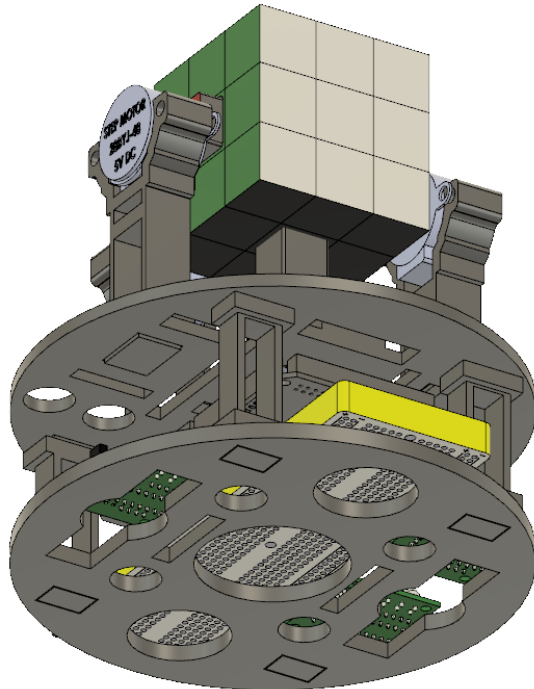
- Records (s_t, a_t, s_{t+1})
- Executes action sequences
- Timestamps all transitions

PHYSICAL SYSTEM: Rubik's Cube (state $s \in S$)

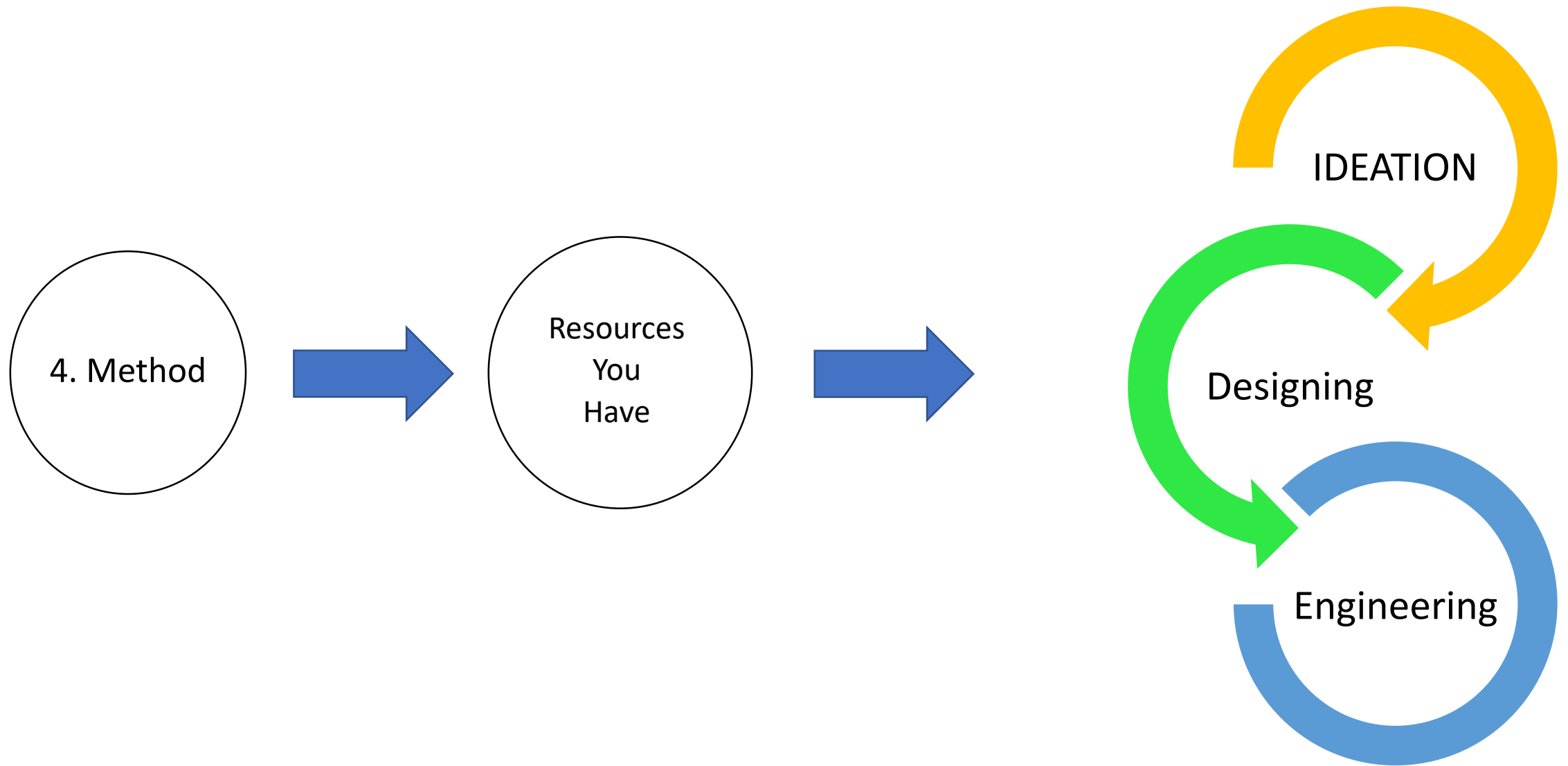


3. Solution

- Our approach leverages 3D printing technology to create durable and customizable parts, combined with readily available electronic components. This enhances the feasibility of replicating the robot in educational settings while providing insights into automation and robotics.



Total Cost: only 400-500 RMB	
Component	Estimated Cost (RMB)
3D Printed Parts (PLA)	~50
Arduino Mega 2560	~120
Stepper Motors (28BYJ-48)x3	~60
ULN2003 Drivers	~30
Power Supply & Cables	~40
Miscellaneous (Screws, Teflon, etc.)	~50
Total	~350-400 RMB



5. Design Fabricating

Sketch

Design and Simulation

Fabrication Prototyping

Assembly

Testing

Analysis

Reporting

Software's



Tools



Universal Laser Cutter



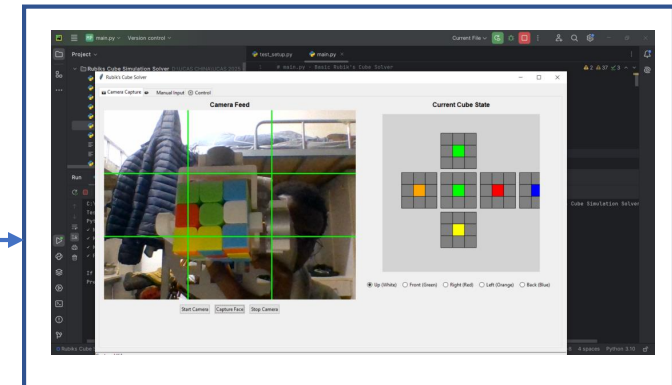
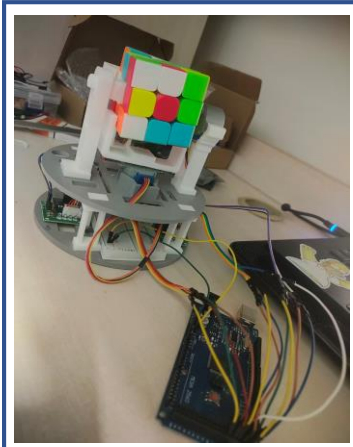
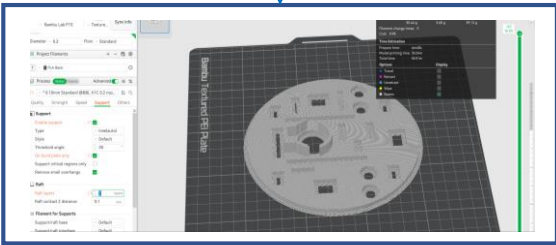
3D Printer



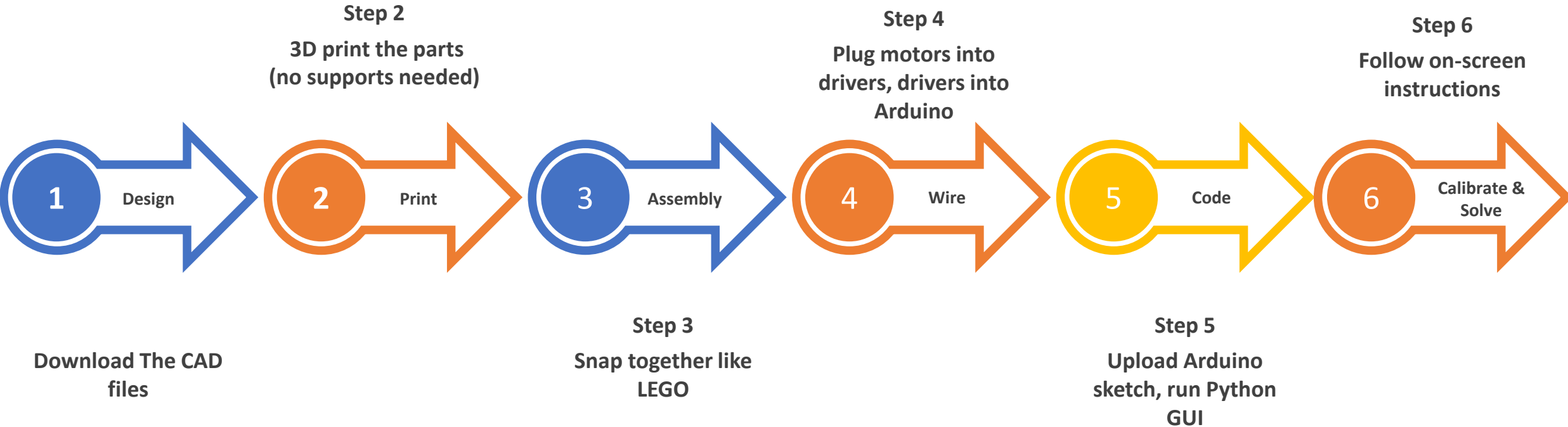
SLA Printer

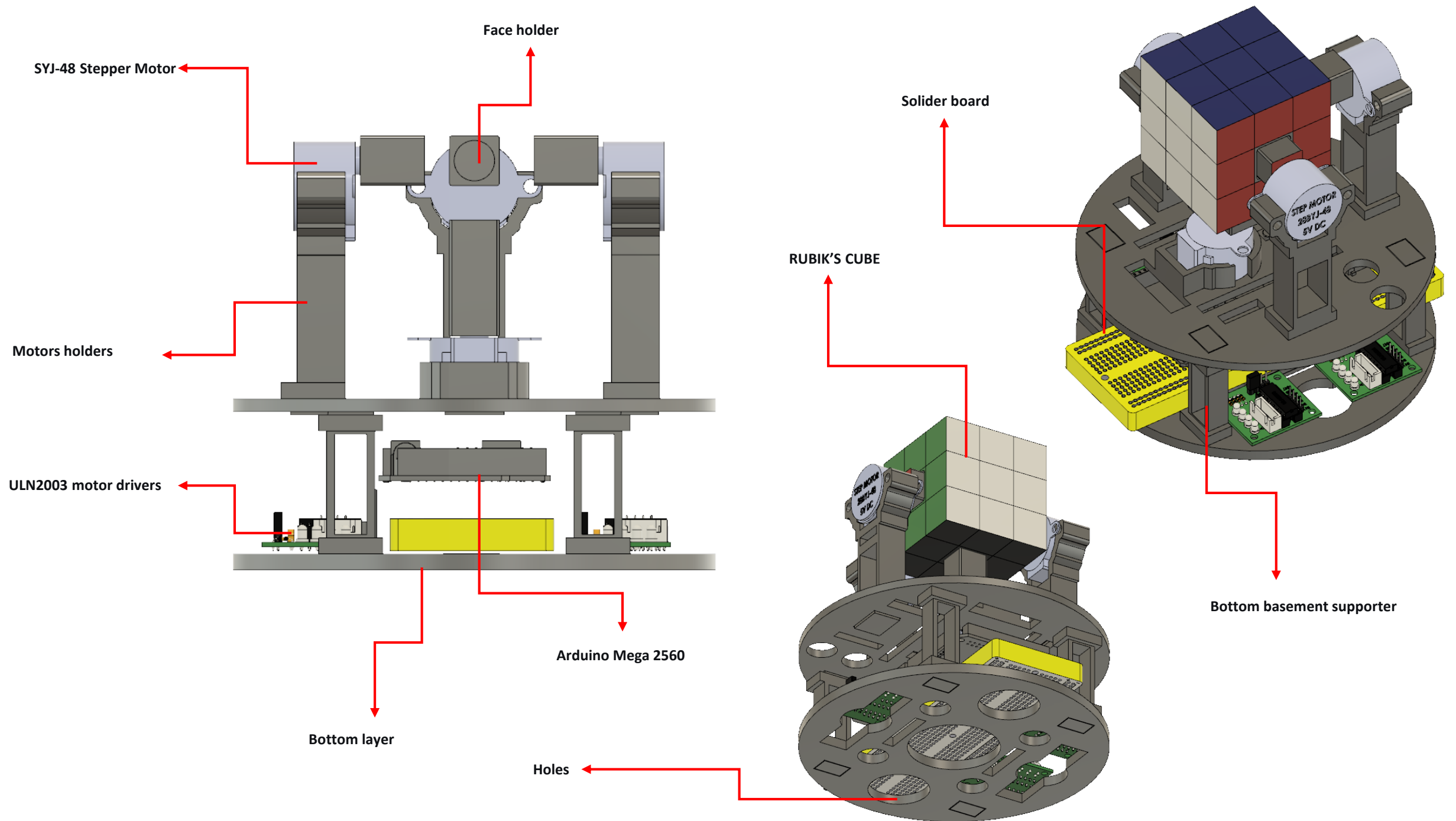


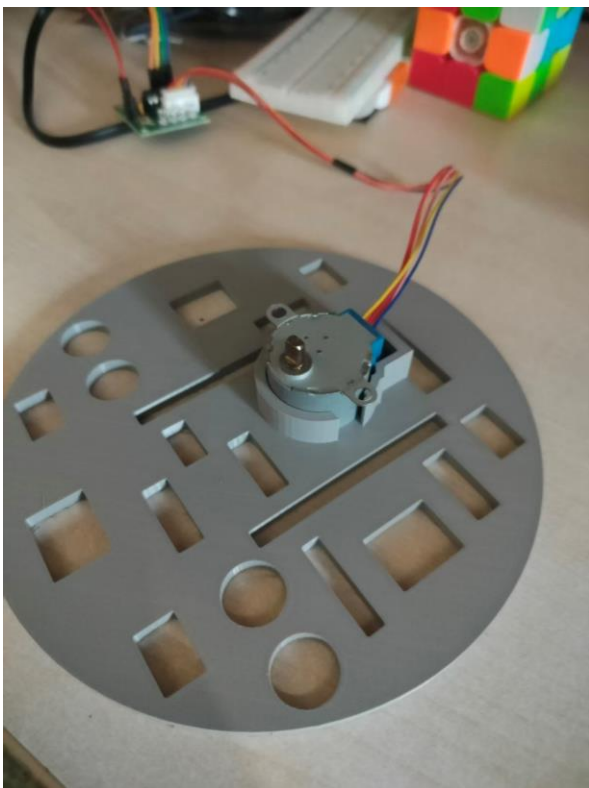
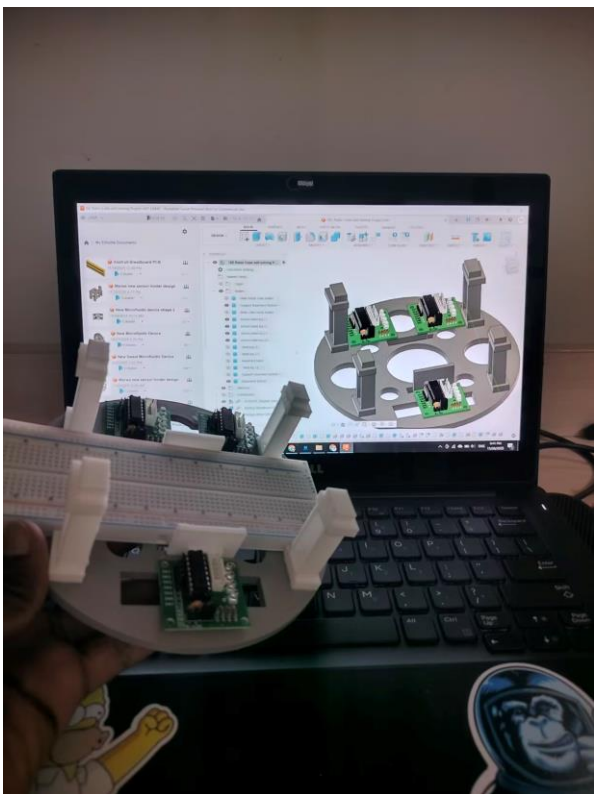
Micro-milling Machine

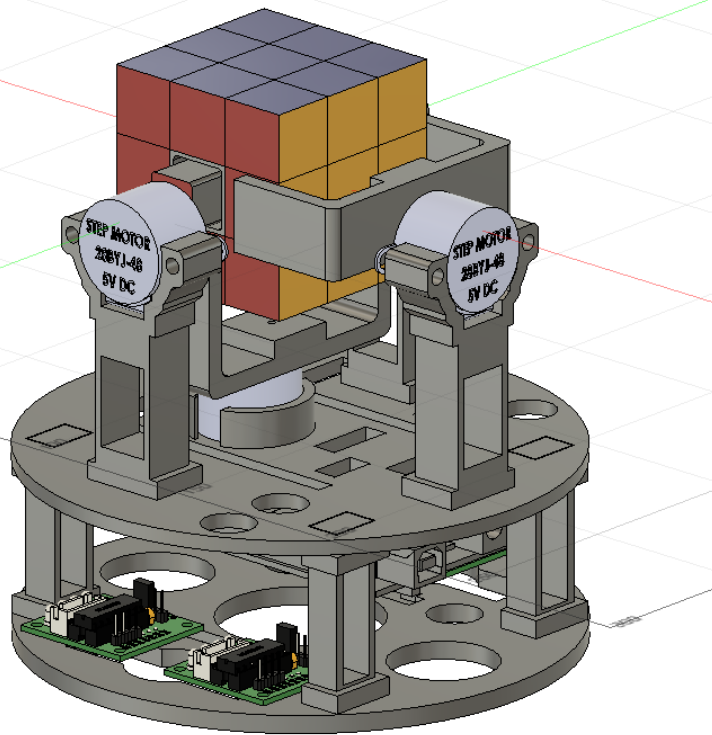


Build in 6 Easy Steps









6. Challenges & Problems

1. What is the difference between those two models ?
2. Why did we changed it?
3. IS the first design good ?
4. How much time it take to change to this new model ?
5. What could be the challenges, if we strike to the first design ?
6. What we learned from this ?

Key Challenges:

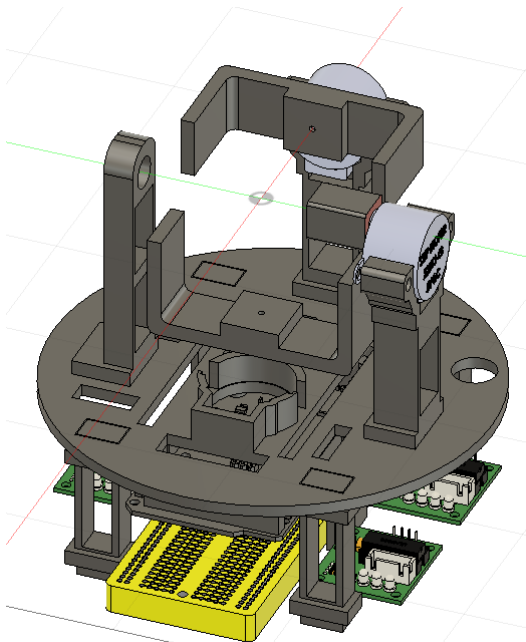
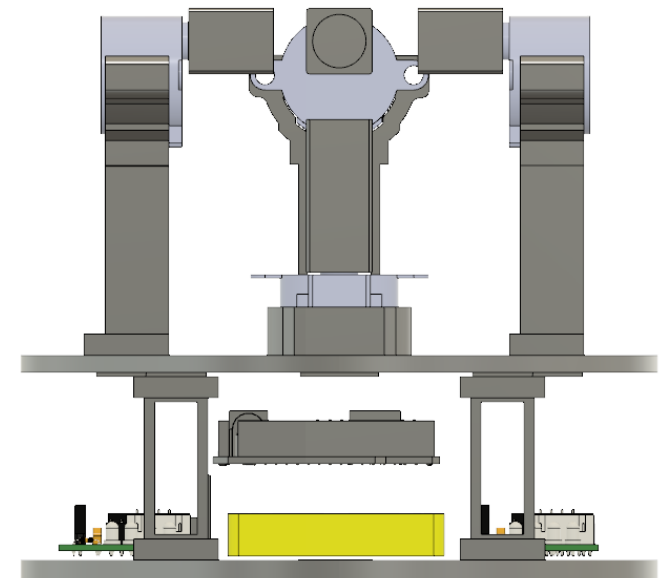
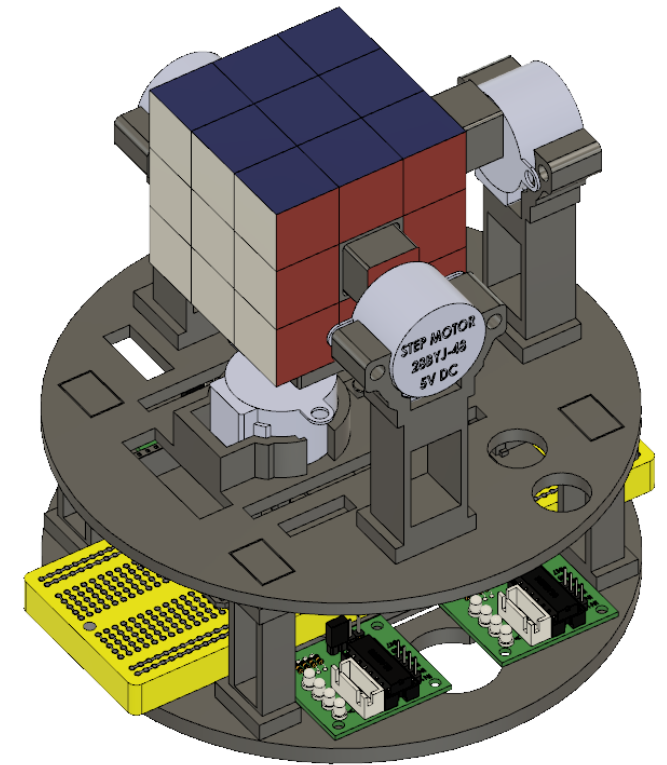
1. Mechanical stability with limited components
2. Accurate rotation with low-torque motors
3. Color detection without high-end vision systems

Mechanical Challenges:

1. Tolerance issues with 3D printing
2. Motor torque limitations
3. Cumulative alignment errors

Solutions Implemented:

1. Iterative CAD redesigns
2. Software position recalibration
3. External power supply addition



Electronics Implementation (Control System Components):

Hardware Components:

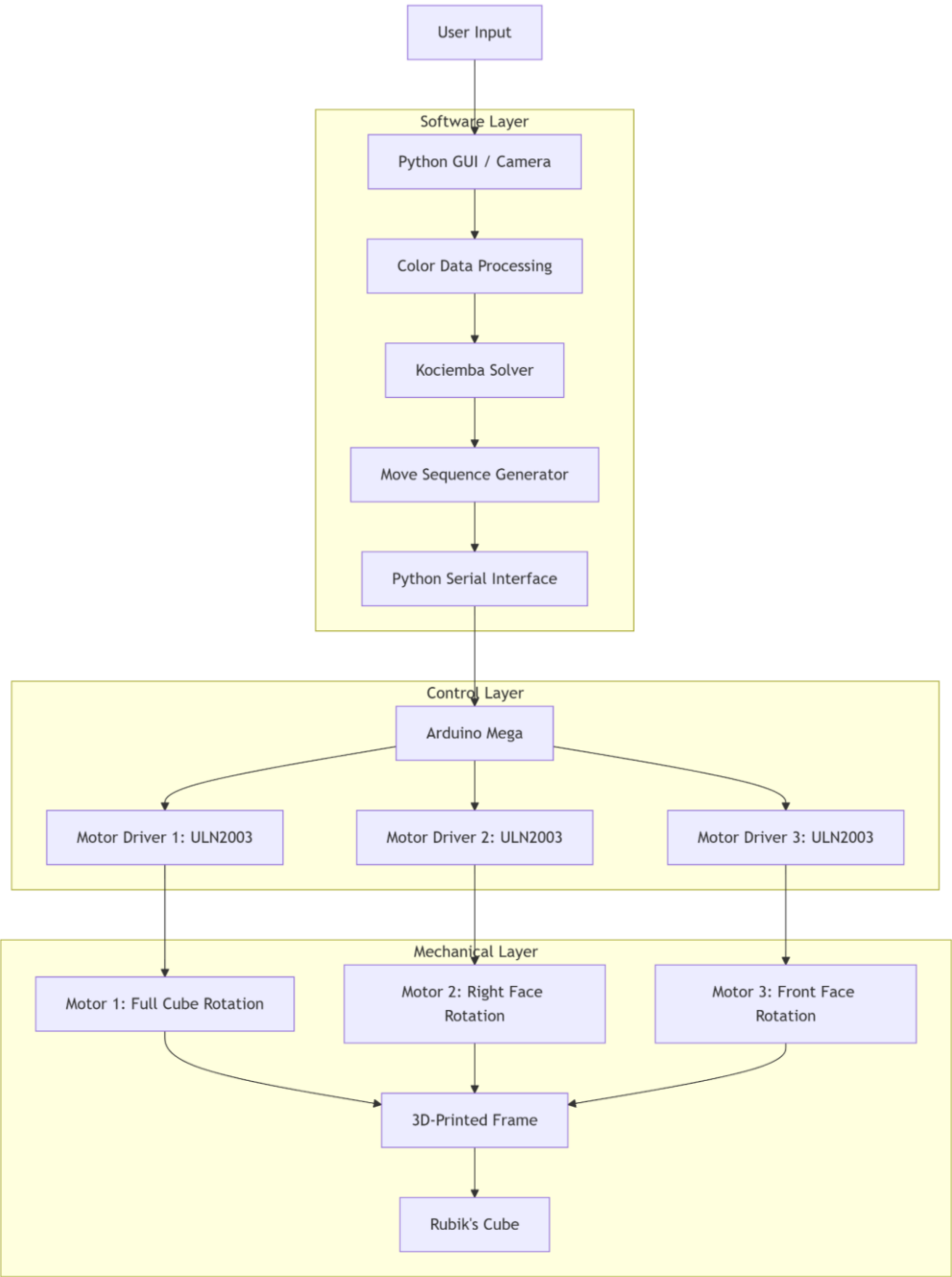
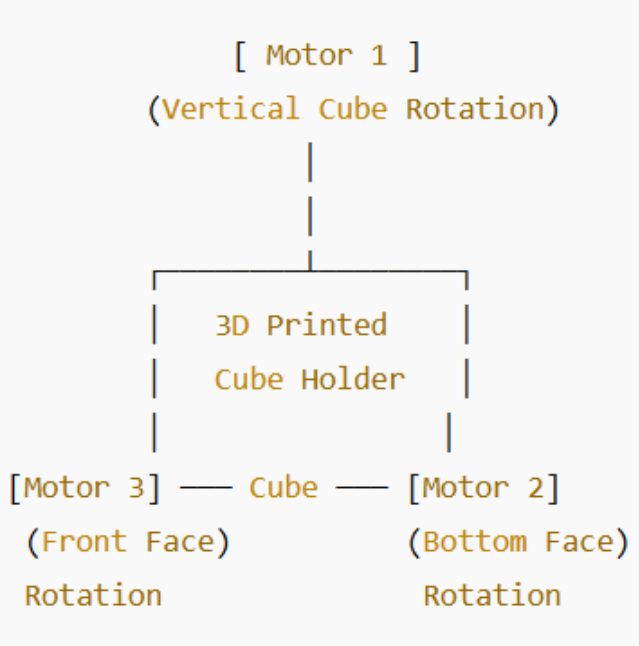
- 1. Arduino Mega 2560 (Main controller)
- 2. 28BYJ-48 Stepper Motors (3 units)
- 3. ULN2003 Driver Boards
- 4. External 5V, 2A Power Supply

Key Specifications:

- 1. Motor torque: 350 gf·cm each
- 2. Step resolution: 512 steps/90° rotation
- 3. Serial communication: 9600 baud rate

Modular Input Options:

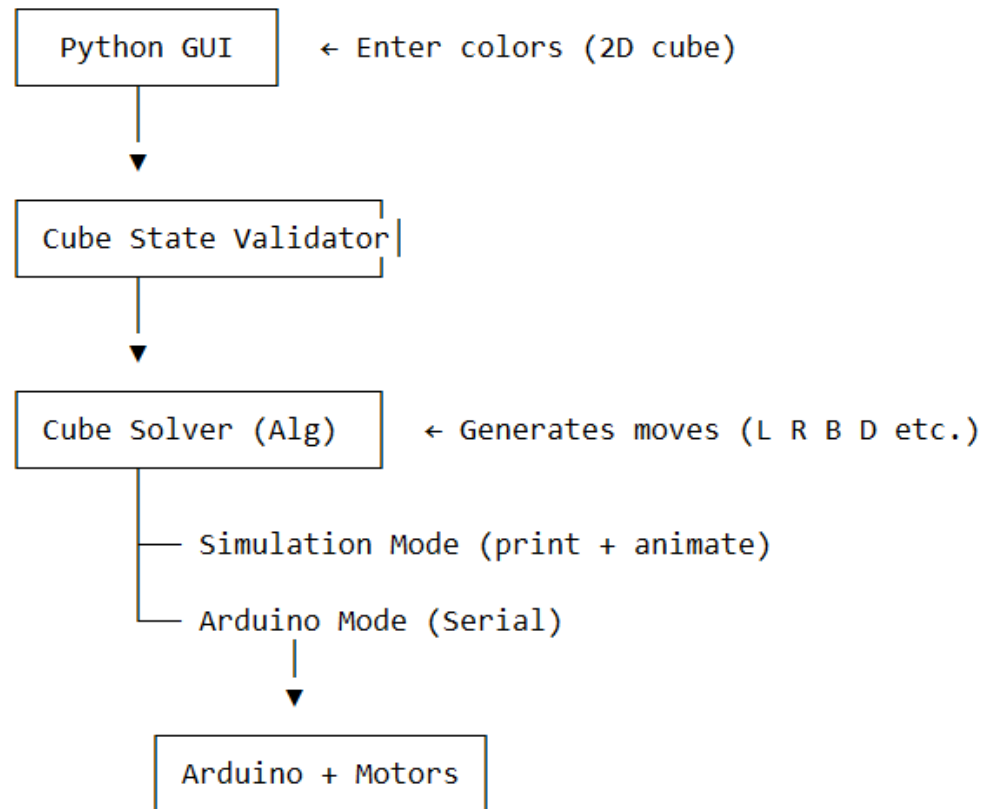
- 1. Color sensor (TCS3200)
- 2. Manual GUI input
- 3. Bluetooth camera interface



7. Algorithms

Software Layer:

1. Python GUI for color input
2. Kociemba solving algorithm
3. Serial communication interface



Software Challenges:

1. Timing synchronization
2. Move sequence optimization
3. Color detection accuracy

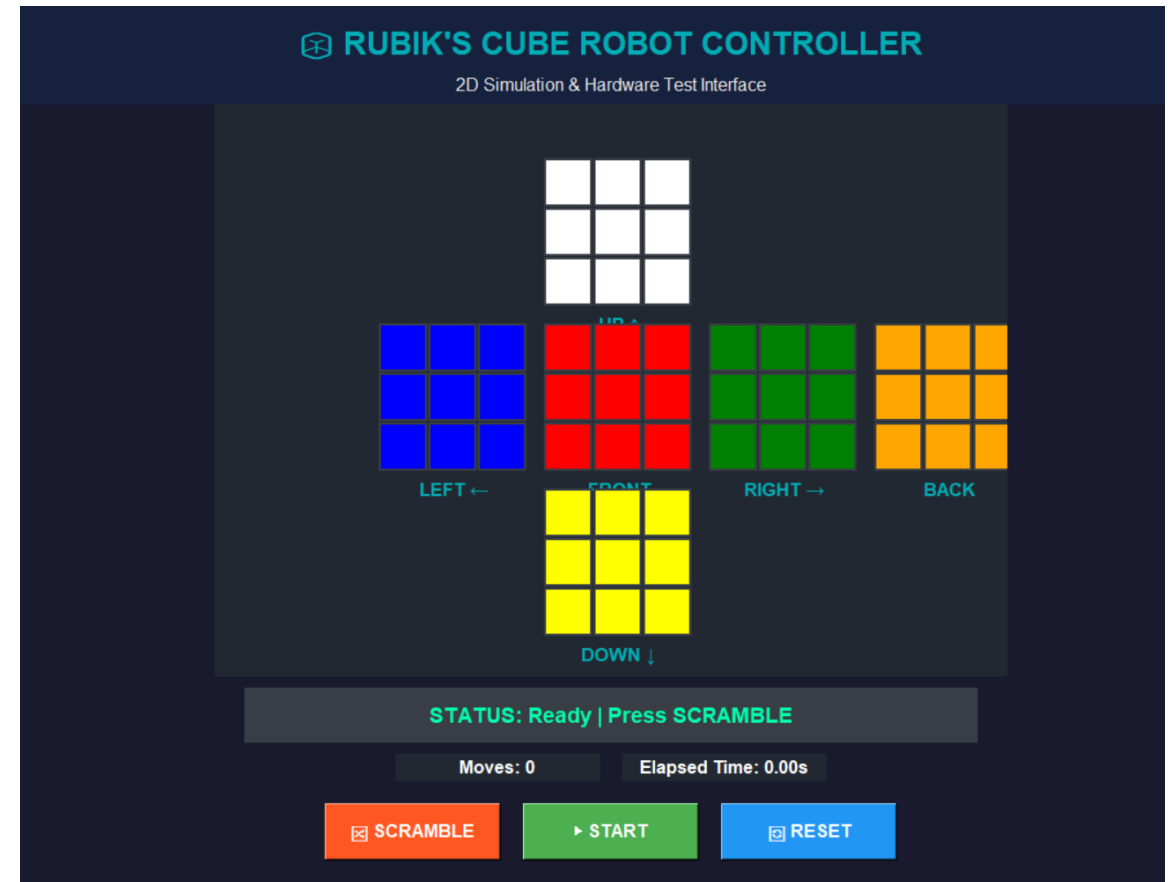


Fig. Shows GUI and algorithm architecture

Fig. GUI Control interface

Can This Algorithm Solve Any Scramble?

No, this algorithm cannot solve arbitrary Rubik's Cube scrambles. It is not a full Rubik's Cube solver.



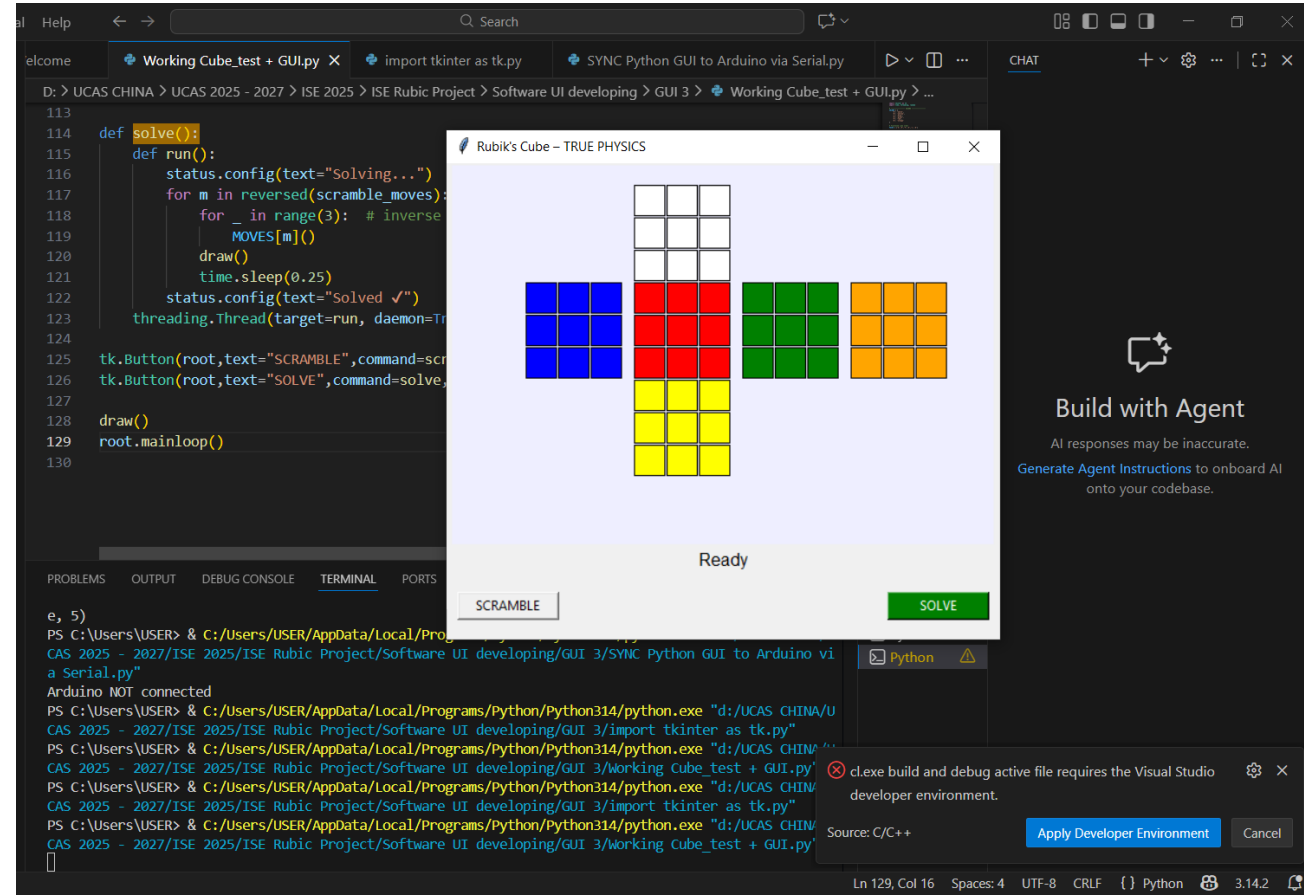
✓ What This Algorithm CAN DO:

- Correctly reverse robot-generated scrambles
- The cube is scrambled only using known moves (U, R, F)
- The exact scramble sequence is stored
- Solving is done by reversing those moves
- This guarantees a correct solution 100% of the time

✓ What This Algorithm CAN DO:

- Cannot solve an unknown or hand-scrambled cube
- Cannot solve complex or arbitrary scrambles
- Cannot use camera input
- Not a general Rubik's Cube solver

This algorithm is intentionally designed for controlled robotic testing rather than full autonomous cube solving. It guarantees correctness under known scramble conditions and provides a reliable foundation for hardware validation and future intelligent extension





How It Works (Simple Version)



Step 1

You can simply impress your audience and add a unique zing and appeal to your Presentations. Easy to change colors, photos and Text.



Step 2

You can simply impress your audience and add a unique zing and appeal to your Presentations. Easy to change colors, photos and Text.



Step 3

You can simply impress your audience and add a unique zing and appeal to your Presentations. Easy to change colors, photos and Text.



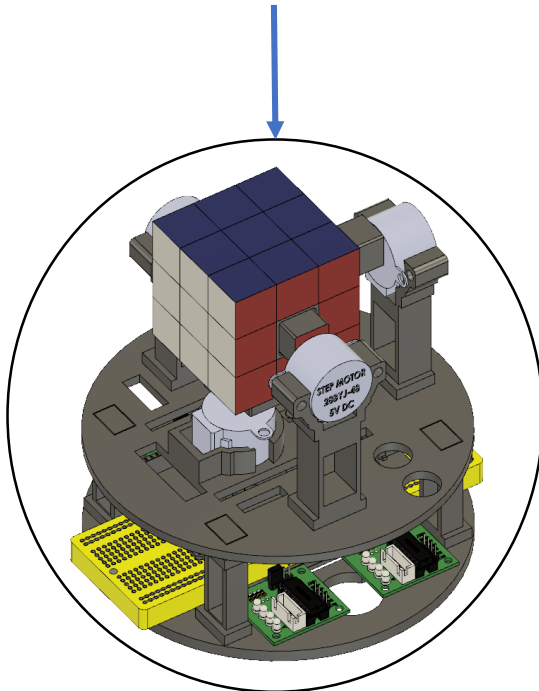
Step 4

You can simply impress your audience and add a unique zing and appeal to your Presentations. Easy to change colors, photos and Text.

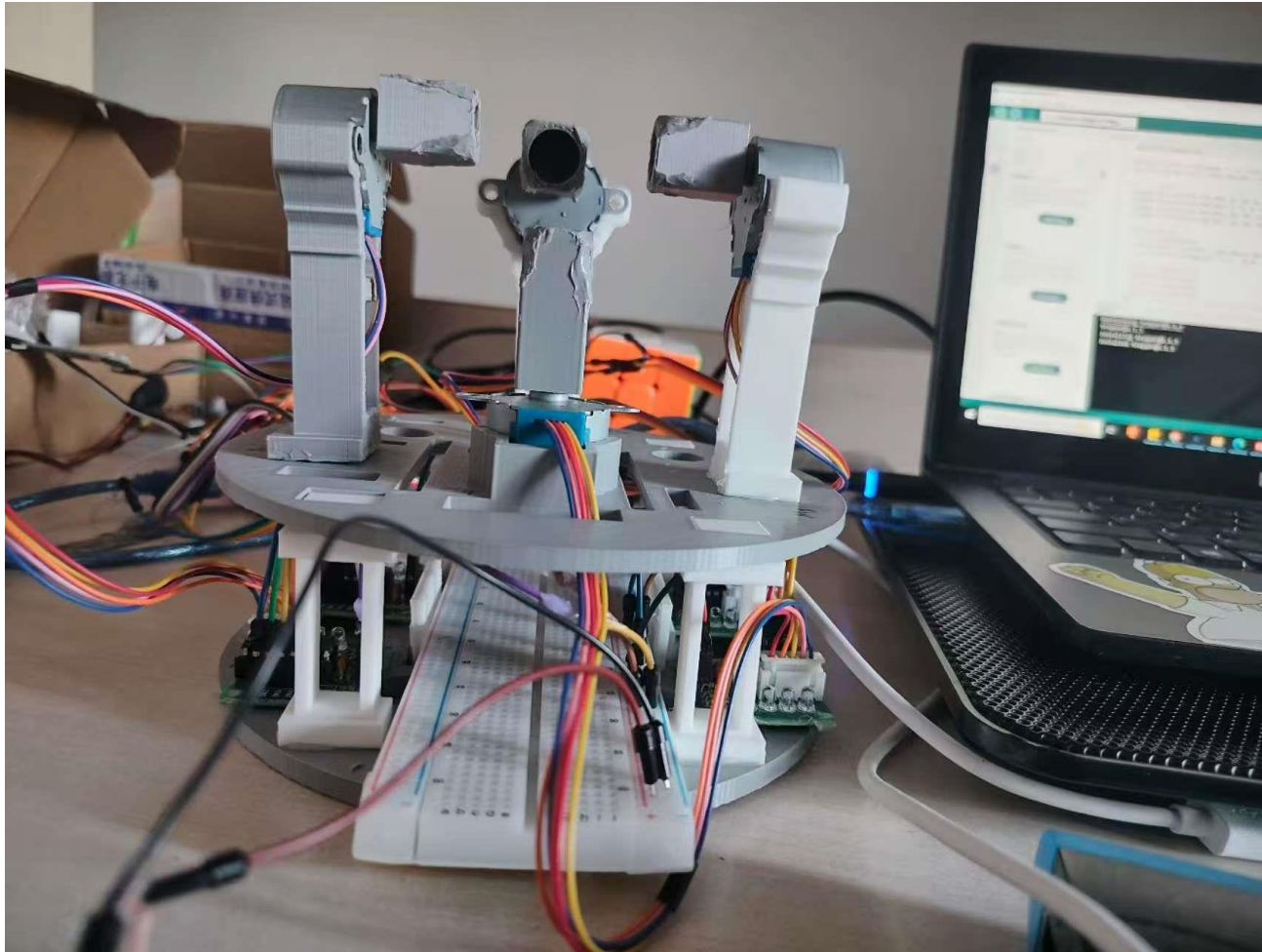


Step 5

You can simply impress your audience and add a unique zing and appeal to your Presentations. Easy to change colors, photos and Text.



8. Contributions



State-of-the-Art Contributions...

Simplicity

Adaptive Sensing Methods

Algorithm Development

Educational Applications

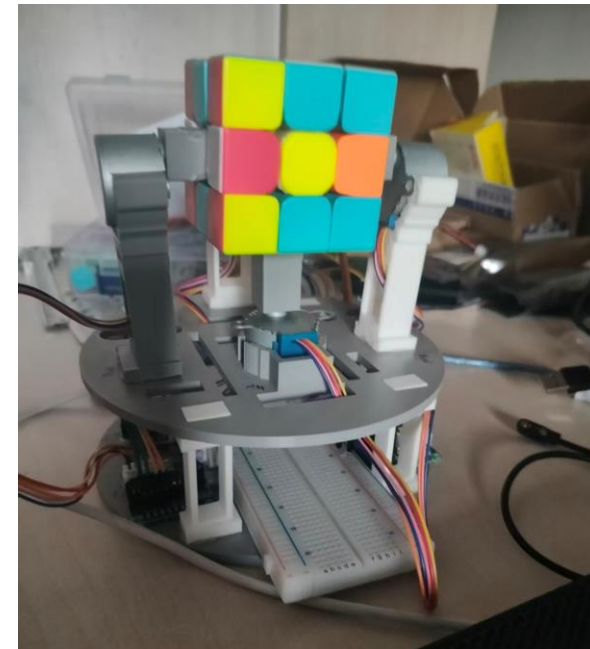
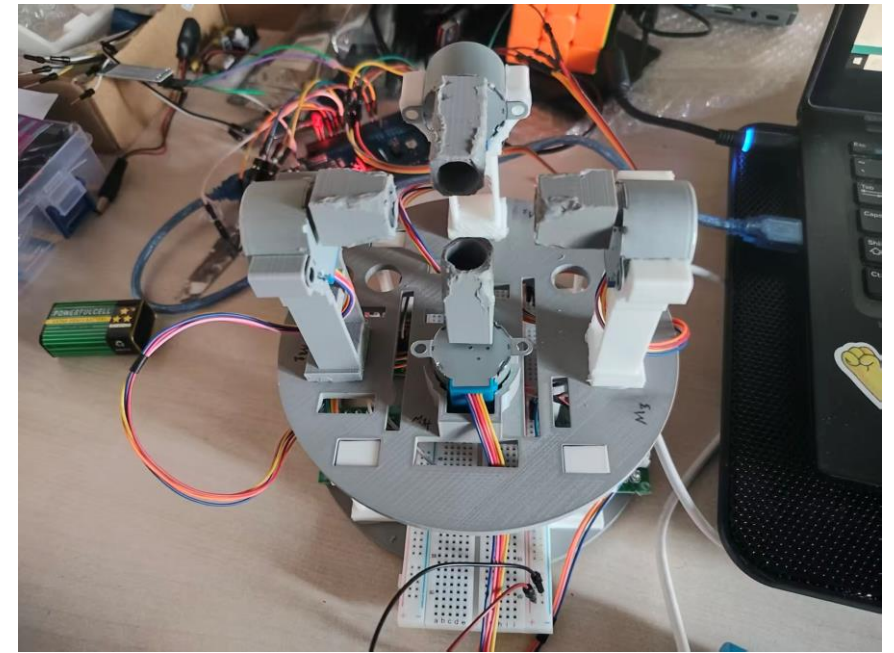
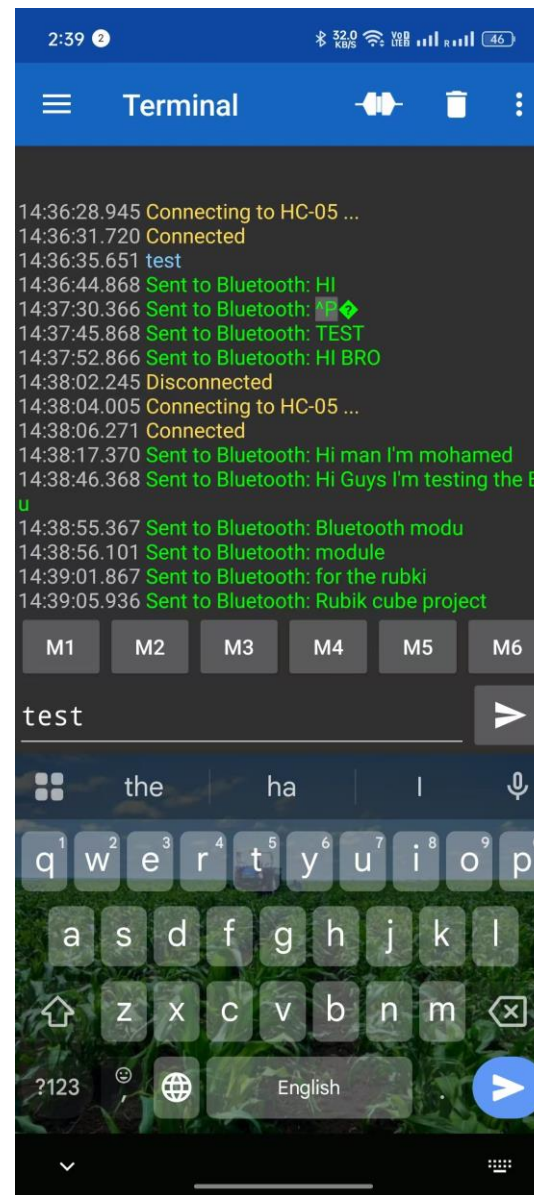
9. Testing and Results

Testing & Results

Testing:

- Motor alignment verification
- Friction reduction with Teflon tape
- Vibration minimization
- Communication stability at 9600 baud

Video (from your laptop)



System Testing and data transmission

10. Applications

Educational Tools: It serves as a practical example for teaching robotics, programming, and STEM concepts.

Robotics Development: The project can be a foundation for more complex robotic systems, encouraging innovation in automation.

Puzzle Solving: Beyond education, the robot can be used in competitions or demonstrations, enhancing interest in robotics and problem-solving.

Research Platform: It offers a platform for testing various algorithms and sensory technologies in robotic applications.

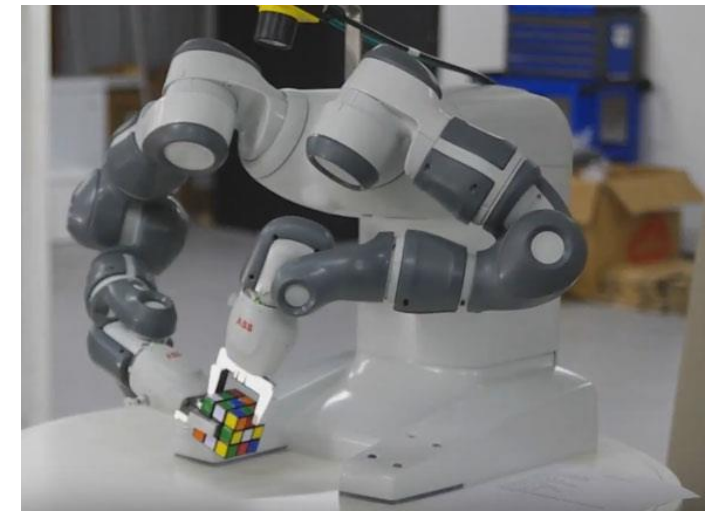
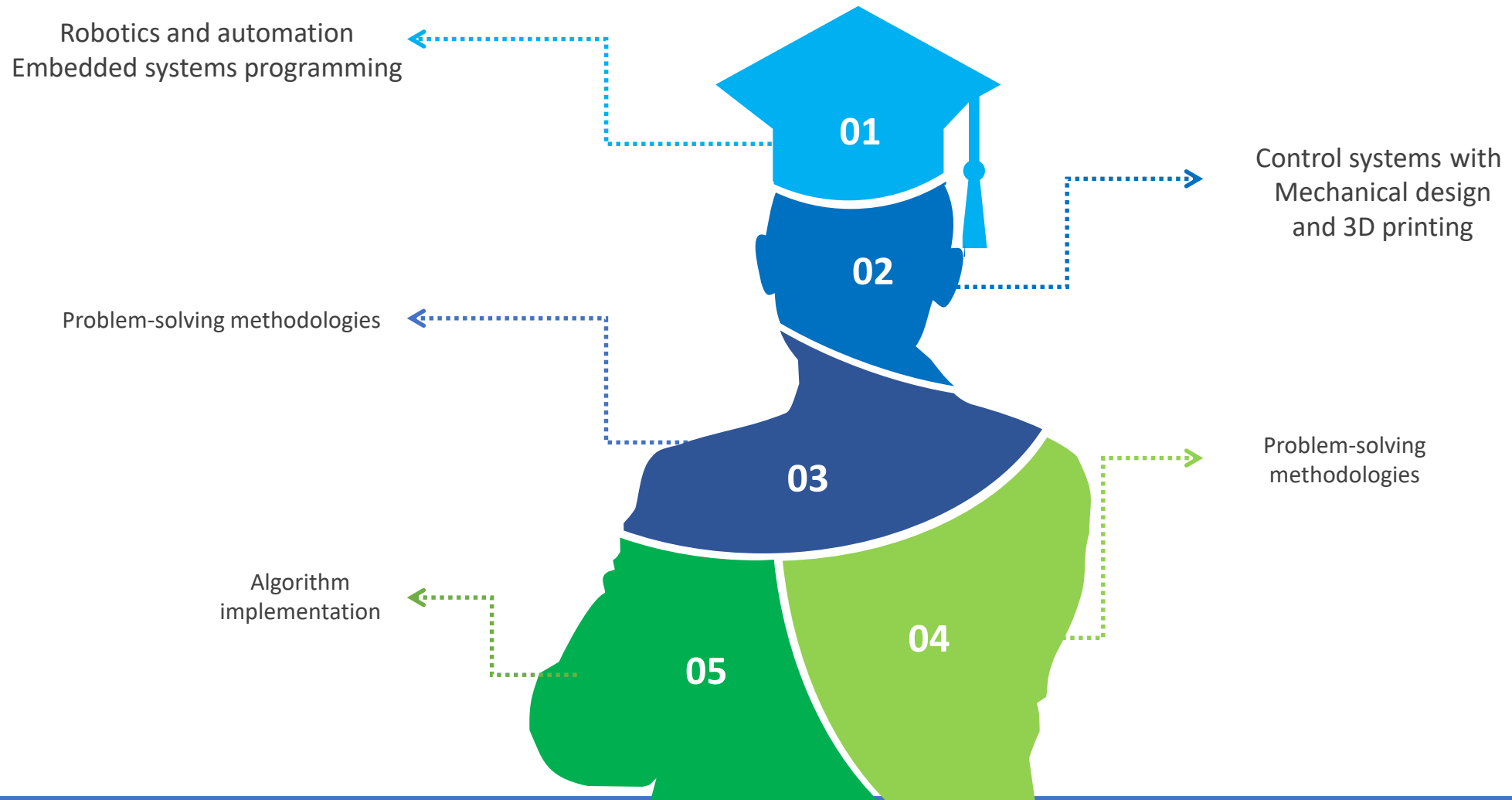


Fig. [YuMi Robot Solving Rubik's Cube - Robotic Gizmos](#)



Learning Outputs



12. Future works

Our approach leverages 3D printing technology to create durable and customizable parts, combined with readily available electronic components. This enhances the feasibility of replicating the robot in educational settings while providing insights into automation and robotics.

Changed and replaced the motors with more high torque and Develop mobile app interface.

Implement reliable camera-based scanning and Improve mechanical rigidity with PETG/ABS

Add automated calibration routines and make the robot Fully autonomous operation

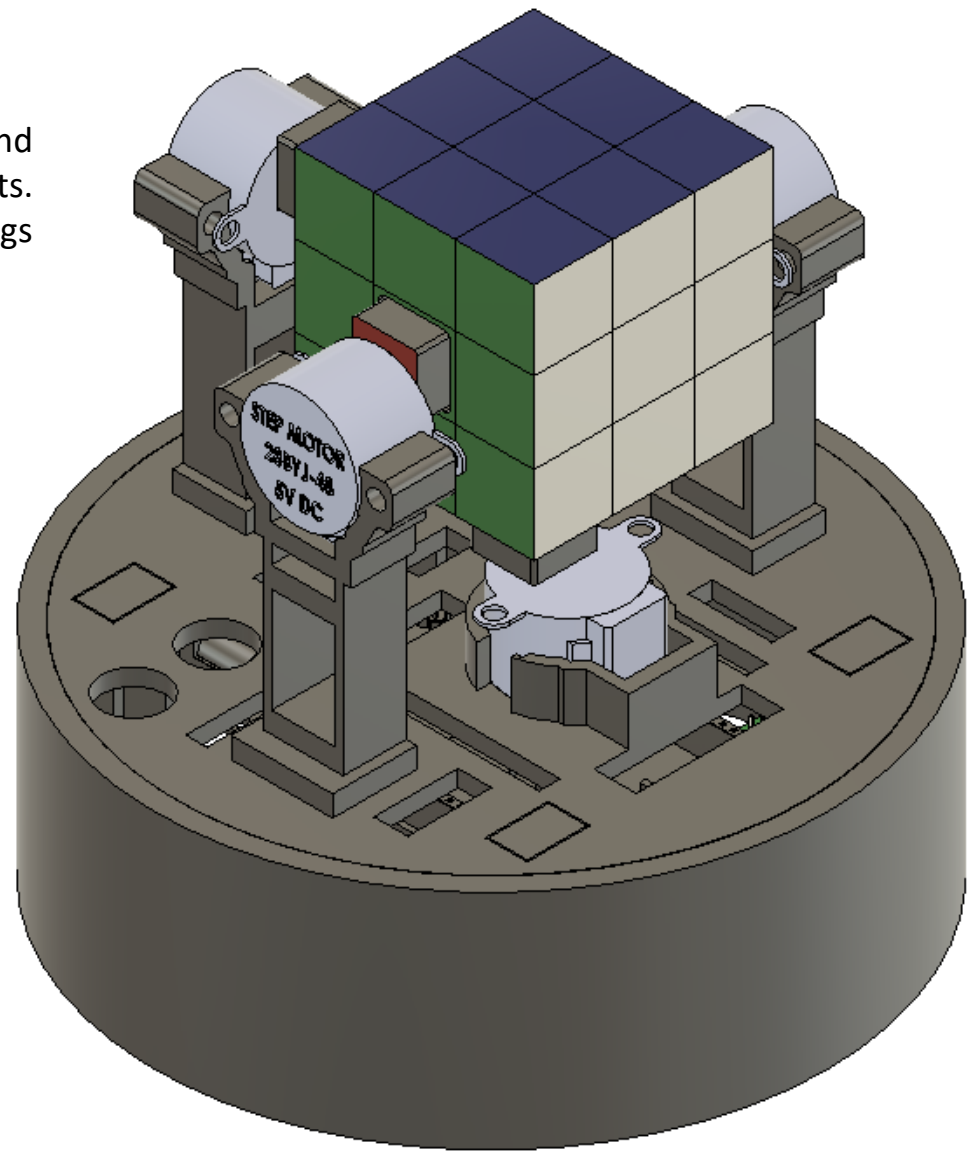
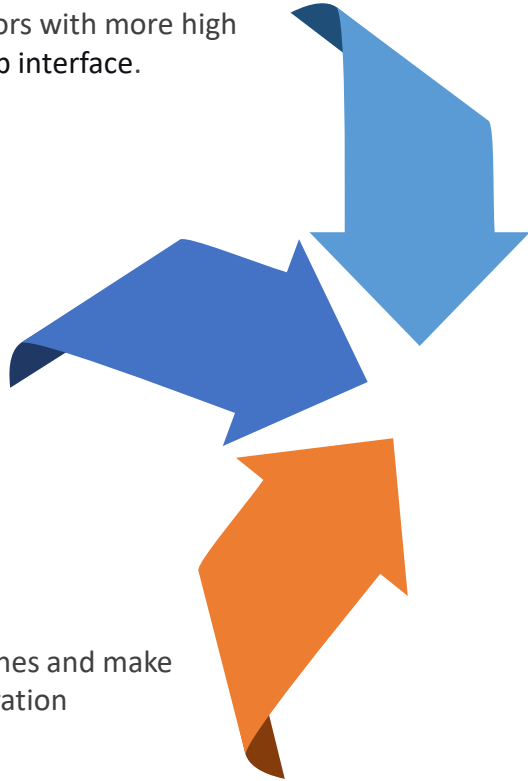
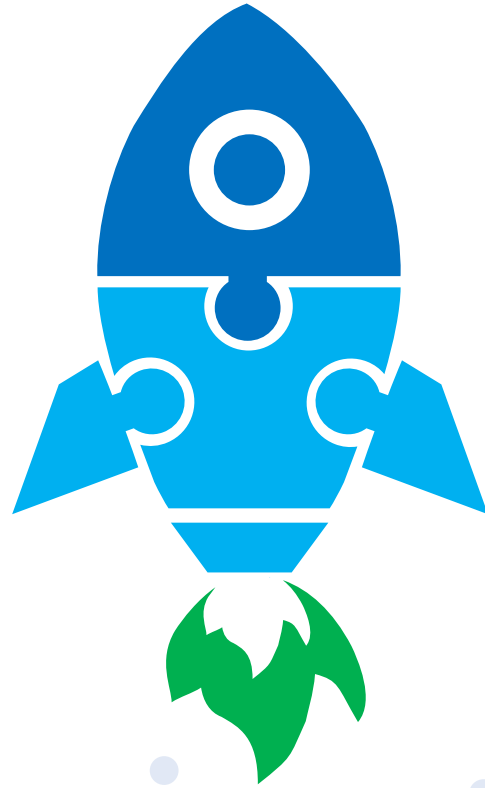


Fig. updated design for the Cube Robot's



“Try It Yourself – No
Experience **Needed!**”

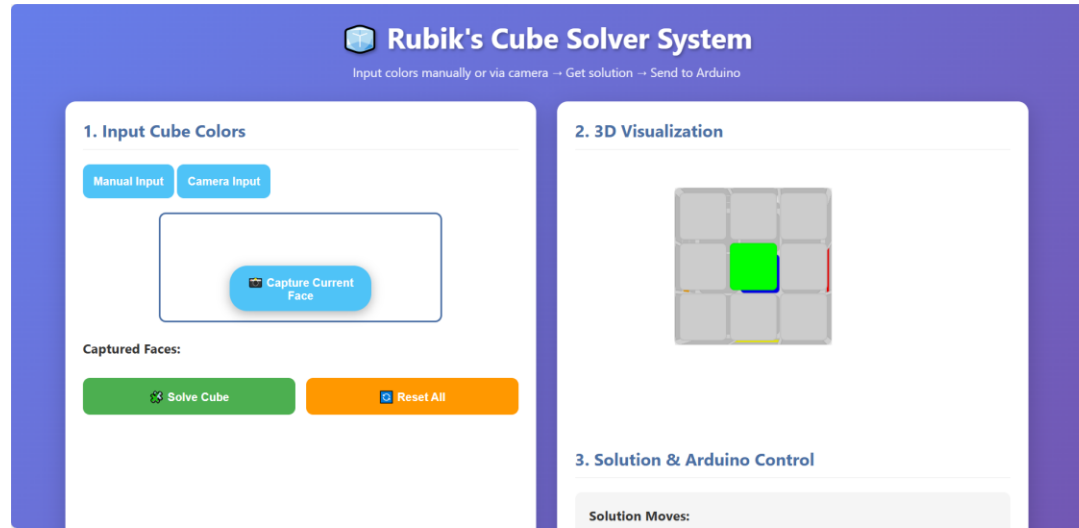


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- Video tutorial available
- All files open-source
- Online community support
- Perfect for school projects

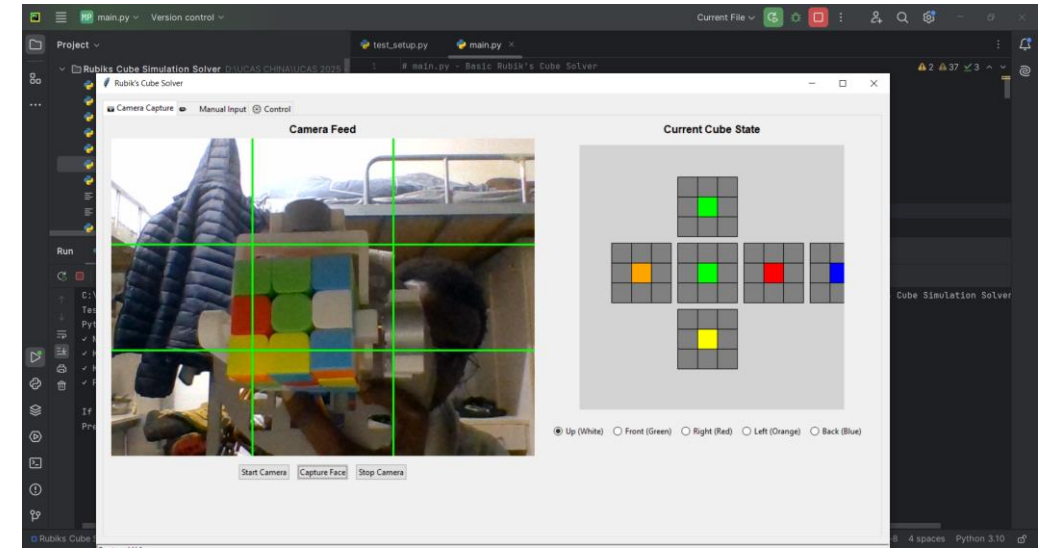
Future Improvements

Develop and testing different GUI Systems and algorithms



Rubik's Cube Solver System

Input colors manually or via camera → Get solution → Send to Arduino



Rubik's Cube Solver System

Input colors manually or via camera → Get solution → Send to Arduino

“Try It Yourself – And
Build Your own **ROBOT!**”

GitHub Repositories:

<https://github.com/0MohamedGama0/3D-Printed-Self-Solving-Rubiks-Cube-Robot>